ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

- **English Name: HSINYU YEH**
- **Chinese Name: 葉信妤**
- **Email: jollyyeh@gapp.nthu.edu.tw**
- **Student ID: 114071516**
- **University: National Tsing Hua University**

Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594

2. Referring to the NUS definition of Frequent Pattern Mining, if we separate the type of data we want to analyze, we can avoid other types of data becoming noise, allowing algorithms to better represent the knowledge of a specific domain.

AI used for this answer: Yes

If yes, how: Verifying the problem requirements and check the code I wrote myself or my thoughts on the answer.

External references (links):

https://www.comp.nus.edu.sg/~ashishd/IT5006/notes/35-fpmining/

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: Dropping 40% of the documents will lose a lot of available text data and may lead to a decline in the representativeness of the remaining dataset compared to the original data. An alternative is to use KNN (K-nearest neighbor) classification method to estimate the missing category labels according to similar documents.

AI used for this answer: Yes

If yes, how: Discuss the answer I proposed and search for information.

External references (links):

https://medium.com/@tarangds/knn-imputation-the-complete-guide-146f932870a7

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. No, the category distribution of the sample is roughly similar to the original one, but not exactly the same.

2. Some categories may be sampled too sparsely, making the sample not representative of the original data well.

AI used for this answer: No

If yes, how:

External references (links): From Professor Chen's handout and lessons.

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer: 1. The number of features increased from 35,482 to 35,788

2. Adding stop words will reduce the sparsity of the matrix, because words like "the", "a" and "is" appear frequently across many documents, resulting in more none-zero elements in the matrix.

AI used for this answer: Yes

If yes, how: Discuss the answer I proposed and search for information.

External references (links):

https://scikit-learn.org/stable/modules/feature_extraction.html

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. As the variance threshold increase from 0 to 0.05, the number of retained words decreased, because the higher the threshold, the more features with low variation will be removed.

2. The first to be removed are the words with low variance, because the document frequency is very low and rare words only appear in a few files. These words are 0 in most documents, so the variance between different documents is very close to 0.

3. Because these features remain almost unchanged across different documents and provide limited distinguishing information, removing them can reduce data dimensionality while typically not losing much useful information for model judgments.

Threshold	**Number of terms kept (keep == True)**
0.0	35,482
0.01	5,089
0.05	1,491

AI used for this answer: Yes

If yes, how: Confirming the results of the code I ran, I shared my thoughts on the answer and reference for the answer, and then discuss the answer with AI.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.VarianceThreshold.html

https://scikit-learn.org/stable/modules/feature_selection.html

---

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what Pearson correlation measures versus what Spearman rank correlation measures?
3. In what specific scenario would a word have a high Spearman rank correlation with a topic, but a significantly lower Pearson correlation?

Answer:

1. Pearson's results tend to be general words, with only some words related to graphics; Spearman's findings were more distinct, as terms like graphics, files, image and 3D were clearly connected to the topic, so the words found by Spearman are more in line with the domain-specific terminology of graphics.

2. Pearson correlation measures the strength and direction of a linear relationship between the original values, while Spearman correlation measures the strength and direction of a monotonic relationship based on rank.

3. If a word appears more often as the topic becomes more relevant, but the increase is not linear, its Spearman correlation can still be high because the ranking remains consistent, while its Pearson correlation maybe lower because the original values do not follow a linear relationship.

AI used for this answer: Yes

If yes, how: Help to understand the definitions and formulas of two kinds of correlation, and discuss whether the answer I provide is relevant to the question.

External references (links):

https://www.surveymonkey.com/learn/market-research/pearson-correlation-vs-spearman-correlation/

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

When minSup deceases from 9 to 3, the number of patterns increases a lot, from 710 to 9,547, while the runtime also increases from about 0.0818 seconds to 2.6448 seconds. A lower minSup keeps more patterns because fewer occurrences are required.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

K controls the number of patterns kept, while minSup controls the minimum frequency required for a pattern to be kept.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent patterns are more concise because they keep only the largest frequent itemset and remove smaller frequent patterns that are subsets of them.

AI used for this answer: Yes

If yes, how: Analyze the question, introduce Frequent Pattern Mining, check the code and final result.

External references (links):

https://hanj.cs.illinois.edu/cs412/bk3/06.pdf

https://www.researchgate.net/publication/4285466_Mining_Top-Rank-K_Frequent_Patterns

https://en.wikipedia.org/wiki/K-optimal_pattern_discovery

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. X_counts contains tens of thousands of features, so it is too high-dimensional to plot directly on a 2D scatter plot.

2. We need to apply dimensionality reduction first, such as PCA, t-SNE or UMAP to reduce the data to 2D or 3D.

AI used for this answer: Yes

If yes, how: Analyze the contents of the chart, introduce PCA, t-SNE, UMAP and other methods, discuss and check answers.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html

https://experiments.withgoogle.com/visualizing-high-dimensional-space

---

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. UMAP shows the most distinct and separated cluster structures among the four categories, while PCA shows heavy overlap with most observations compressed around the center.

2. PCA is a linear method, so it may not capture complex non-linear relationships in high-dimensional text dat. t-SNE and UMAP can model non-linear neighborhood structures, so they often separate categories more clearly.

3. Because t-SNE mainly preserves local neighborhood relationships rather than global distances, the distance between clusters in the 2D plot may be distorted and does not necessarily reflect the true distance in the original high-dimensional space.

AI used for this answer: Yes

If yes, how: Clarified the difference between local and global distances in t-SNE, check and discuss my original answer.

External references (links):

https://distill.pub/2016/misread-tsne/

https://jtr13.github.io/cc21fall1/efficient-dimension-reduction-with-umap.html

---

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. Four binary columns are created, one for each category. For instance [1,0,0,0], [0,1,0,0], [0,0,1,0] and [0,0,0,1].

2. The algorithm may incorrectly assume that the categories have an order and meaningful numerical order and distance. For example, it may treat category 3 as larger and farther from category 0 than category1, even though these numbers are only labels.

3. One-hot binarization gives each category its own binary position, so the model no longer treats category labels as ordered numbers or assumes that some categories are farther apart than others.

AI used for this answer: Yes

If yes, how: Introduce the concept of one-hot encoding, refine my answer and verify it with an external reference.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.OneHotEncoder.html

https://www.digitalocean.com/community/tutorials/understanding-one-hot-encoding-in-machine-learning

---